

DEL BONITA AGDM, AB, Canada

WMO: 712240

Lat: 49.0500N

Lon: 112.8099W

Elev: 1310

StdP: 86.55

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.0	-23.9	-30.5	0.3	-26.6	-27.2	0.4	-23.7	16.9	4.2	14.7	1.2	3.9	320	0.773

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	30.1	14.5	28.2	14.1	26.2	13.7	16.6	25.1	15.6	24.5	14.7	23.8	4.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	11.6	19.2	12.5	10.6	18.0	11.2	9.7	17.1	51.6	25.2	48.4	24.3	45.7	23.9	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	12.6	11.1	-30.9	33.2	4.1	2.0	-33.9	34.6	-36.3	35.7	-38.6	36.8	-41.5	38.3		
(5)			-31.1	18.9	4.0	1.1	-34.0	19.6	-36.4	20.2	-38.7	20.8	-41.6	21.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-5.2	-6.3	-1.3	3.5	8.7	12.7	17.0	16.3	11.4	5.0	-1.4	-6.3	(6)
(7)		DBStd	10.16	7.72	7.96	6.64	4.97	4.46	3.38	3.29	3.68	4.94	5.71	7.15	7.76	(7)
(8)		HDD10.0	2631	471	458	349	200	79	11	1	3	41	171	343	504	(8)
(9)		HDD18.3	5071	729	691	607	445	300	172	64	82	212	414	592	762	(9)
(10)		CDD10.0	652	0	0	1	5	38	92	218	199	83	15	2	0	(10)
(11)		CDD18.3	51	0	0	0	0	1	3	23	20	4	0	0	0	(11)
(12)		CDH23.3	1181	0	0	0	1	19	65	513	448	131	4	0	0	(12)
(13)		CDH26.7	339	0	0	0	0	1	11	166	132	29	0	0	0	(13)
(14)	Wind	WSAvg	5.3	6.3	5.5	5.8	5.6	5.2	5.0	4.2	4.2	4.8	5.5	6.1	6.0	(14)
(15)	Precipitation	PrecAvg	379	12	10	18	31	57	90	40	36	32	22	16	14	(15)
(16)		PrecMax	579	21	22	40	62	106	225	139	83	93	69	34	29	(16)
(17)		PrecMin	204	2	1	4	13	16	15	2	1	3	5	4	1	(17)
(18)		PrecStd	89	5	5	9	12	25	51	31	21	23	15	8	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	11.2	15.3	21.4	25.7	28.4	33.0	29.6	22.6	16.9	9.7	(19)	
(20)			MCWB	4.4	3.7	6.3	9.4	12.9	16.3	15.8	14.4	12.9	10.3	7.5	3.8	(20)
(21)		2%	DB	7.7	7.6	12.1	17.5	22.8	25.1	30.5	29.9	27.2	19.2	12.4	6.7	(21)
(22)			MCWB	3.1	2.5	5.1	7.7	11.3	14.3	15.1	14.0	12.4	9.3	6.1	2.5	(22)
(23)		5%	DB	5.7	5.5	9.6	14.9	20.3	23.0	28.4	28.0	24.3	16.1	9.7	4.9	(23)
(24)			MCWB	1.9	1.4	4.0	6.4	10.2	13.3	15.0	13.6	11.7	8.3	4.3	1.6	(24)
(25)		10%	DB	4.1	3.6	7.4	12.1	17.7	20.8	26.2	25.9	21.2	13.3	7.4	3.3	(25)
(26)			MCWB	1.1	0.4	2.8	5.2	9.2	12.3	14.6	13.2	10.6	6.6	3.0	0.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	4.6	7.1	10.0	13.4	17.5	18.7	16.6	14.6	11.8	8.7	4.8	(27)
(28)			MCDWB	9.4	9.5	14.0	18.5	23.2	25.3	25.7	25.8	23.8	20.1	14.9	8.6	(28)
(29)		2%	WB	3.2	3.0	5.7	8.2	11.8	15.7	17.2	15.5	13.3	9.9	6.3	2.8	(29)
(30)			MCDWB	7.0	7.1	11.0	16.6	21.0	23.2	25.4	24.9	23.8	18.0	12.0	5.9	(30)
(31)		5%	WB	2.1	1.6	4.4	6.9	10.7	14.2	16.2	14.7	12.2	8.5	4.7	1.8	(31)
(32)			MCDWB	5.3	5.3	9.2	14.1	19.0	21.1	24.9	24.7	22.9	15.4	9.2	4.7	(32)
(33)		10%	WB	1.2	0.4	2.9	5.4	9.7	13.0	15.3	13.9	11.1	7.1	3.2	0.5	(33)
(34)			MCDWB	4.1	3.4	7.0	11.5	16.7	19.4	24.4	23.9	20.5	12.6	7.0	3.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	10.4	10.3	11.5	12.1	12.3	15.5	15.4	13.5	11.8	10.0	9.4	(35)
(36)			MCDBR	9.9	11.6	13.1	16.2	16.6	16.0	18.8	19.1	18.9	15.2	11.9	9.8	(36)
(37)			MCWBR	6.7	7.6	7.6	8.2	7.6	7.7	7.5	7.2	7.8	7.6	7.0	6.8	(37)
(38)		5% WB	MCDWB	9.8	11.9	12.4	15.2	14.8	14.5	16.4	16.4	16.8	14.6	11.5	9.7	(38)
(39)			MCWBR	7.0	8.3	7.6	8.1	7.1	7.5	7.6	6.8	7.3	7.6	7.0	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.237	0.246	0.272	0.306	0.317	0.315	0.316	0.334	0.300	0.269	0.247	0.224	(40)	
(41)		taud	2.444	2.469	2.486	2.416	2.443	2.496	2.499	2.420	2.491	2.560	2.513	2.452	(41)	
(42)		Ebn at Noon	859	934	956	946	944	944	937	904	909	893	842	834	(42)	
(43)		Edh at Noon	60	75	88	106	107	103	101	104	87	68	56	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.24	2.24	3.46	4.76	5.67	6.35	7.12	5.71	4.06	2.50	1.38	0.99	(44)	
(45)		RadStd	0.10	0.17	0.31	0.31	0.50	0.50	0.41	0.37	0.33	0.24	0.10	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page